

Customer Churn Analysis

Telecom subscription business | SQL + Python + Power BI | No machine learning

1. Business Objective and Problem

A telecom company sells phone and internet plans on a monthly subscription. Customers can leave at any time, and every customer lost takes recurring revenue with them. Leadership wants to know **who is leaving, when they leave, and what the company can change** to keep them.

Objective: Find the customer segments driving churn, size the revenue at risk, and recommend retention actions, using descriptive analytics only (SQL aggregations, cleaning, and a dashboard).

Questions the analysis answers

- What is the overall churn rate, and how much monthly recurring revenue (MRR) does it cost?
- Which contract types, services, and payment methods have the highest churn?
- At what point in the customer lifecycle do people leave?
- Which single segment should the retention team target first?

2. Dataset

Item	Detail
Name	Telco Customer Churn (IBM sample dataset)
Size	7,043 customers, 21 columns, one row per customer
Direct CSV link	https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv
Kaggle page (login needed)	https://www.kaggle.com/datasets/blatchar/telco-customer-churn
Target column	Churn (Yes/No): did the customer leave in the last month

Key columns: tenure (months with the company), Contract, InternetService, TechSupport, OnlineSecurity, PaymentMethod, MonthlyCharges, TotalCharges, SeniorCitizen, Churn.

3. Data Cleaning in Python (step by step)

Do this first. The SQL and Power BI sections both use the cleaned file. Install once: `pip install pandas`. Save the downloaded CSV next to your script.

Step 1: Load and inspect

```
import pandas as pd
df = pd.read_csv("Telco-Customer-Churn.csv")
print(df.shape)      # (7043, 21)
print(df.dtypes)      # note: TotalCharges is text, not a number
```

Step 2: Find the problem in TotalCharges

The `isnull()` check shows zero missing values, but `TotalCharges` is stored as text, which means some cells are blank strings. Converting to numbers turns them into NaN so we can see them.

```
print(df.isnull().sum().sum())      # 0
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
print(df["TotalCharges"].isnull().sum())      # 11
print(df[df["TotalCharges"].isnull()][["customerID", "tenure"]])
```

Step 3: Fix it with a reason, not a guess

All 11 rows have `tenure = 0`. They signed up this month and have not been billed, so total charges of 0 is correct.

```
df["TotalCharges"] = df["TotalCharges"].fillna(0)
```

Step 4: Check duplicates

```
print(df["customerID"].duplicated().sum())    # 0, so one row per customer
```

Step 5: Rename columns to snake_case

```
df = df.rename(columns={
    "customerID": "customer_id", "SeniorCitizen": "senior_citizen",
    "PhoneService": "phone_service", "MultipleLines": "multiple_lines",
    "InternetService": "internet_service", "OnlineSecurity": "online_security",
    "OnlineBackup": "online_backup", "DeviceProtection": "device_protection",
    "TechSupport": "tech_support", "StreamingTV": "streaming_tv",
    "StreamingMovies": "streaming_movies", "PaperlessBilling": "paperless_billing",
    "PaymentMethod": "payment_method", "MonthlyCharges": "monthly_charges",
    "TotalCharges": "total_charges"})
df.columns = df.columns.str.lower()
```

Step 6: Simplify labels

Columns like tech_support have three values (Yes, No, No internet service). The third is already captured by internet_service, so merging it into No keeps charts clean.

```
svc = ["multiple_lines", "online_security", "online_backup", "device_protection",
       "tech_support", "streaming_tv", "streaming_movies"]
df[svc] = df[svc].replace({"No internet service": "No", "No phone service": "No"})
df["senior_citizen"] = df["senior_citizen"].map({0: "No", 1: "Yes"})
df["payment_method"] = df["payment_method"].str.replace(" (automatic)", "", regex=False)
```

Step 7: Create analysis columns

churn_flag makes churn easy to sum and average. tenure_order lets Power BI sort the tenure groups correctly instead of alphabetically.

```
df["churn_flag"] = (df["churn"] == "Yes").astype(int)
bins = [-1, 12, 24, 48, 72]
labels = ["0-12 months", "13-24 months", "25-48 months", "49-72 months"]
df["tenure_group"] = pd.cut(df["tenure"], bins=bins, labels=labels)
df["tenure_order"] = df["tenure_group"].cat.codes + 1
df["tenure_group"] = df["tenure_group"].astype(str)
```

Step 8: Final check and export

```
print(df.isnull().sum().sum())    # 0
df.to_csv("telco_clean.csv", index=False)
```

Result: 7,043 rows, 24 columns, no missing values, ready for SQL and Power BI.

4. Business Questions in SQL

Load telco_clean.csv into any database (PostgreSQL, MySQL, SQL Server, or SQLite) as a table named telco_churn. These queries use standard SQL and were tested on the dataset. The output shown under each is what you should see.

Q1. How big is the churn problem?

```
SELECT COUNT(*) AS total_customers,
       SUM(churn_flag) AS churned_customers,
       ROUND(100.0 * SUM(churn_flag) / COUNT(*), 1) AS churn_rate_pct,
       ROUND(SUM(monthly_charges), 0) AS total_mrr,
       ROUND(SUM(CASE WHEN churn_flag = 1 THEN monthly_charges ELSE 0 END), 0) AS mrr_lost,
       ROUND(100.0 * SUM(CASE WHEN churn_flag = 1 THEN monthly_charges ELSE 0 END)
             / SUM(monthly_charges), 1) AS mrr_lost_pct
FROM telco_churn;
-- 7,043 customers | 1,869 churned | 26.5% | MRR 456,117 | lost 139,131 | 30.5%
```

Q2. Which contract types churn most?

```
SELECT contract, COUNT(*) AS customers, SUM(churn_flag) AS churned,
       ROUND(100.0 * SUM(churn_flag) / COUNT(*), 1) AS churn_rate_pct
FROM telco_churn
GROUP BY contract
ORDER BY churn_rate_pct DESC;
-- Month-to-month 42.7% | One year 11.3% | Two year 2.8%
```

Q3. When in the lifecycle do customers leave?

```
SELECT tenure_group, COUNT(*) AS customers, SUM(churn_flag) AS churned,
       ROUND(100.0 * SUM(churn_flag) / COUNT(*), 1) AS churn_rate_pct
FROM telco_churn
GROUP BY tenure_group, tenure_order
ORDER BY tenure_order;
-- 0-12: 47.4% | 13-24: 28.7% | 25-48: 20.4% | 49-72: 9.5%
```

Q4. Does the service mix matter? (internet type and tech support)

```
SELECT internet_service, tech_support, COUNT(*) AS customers,
       ROUND(100.0 * SUM(churn_flag) / COUNT(*), 1) AS churn_rate_pct
FROM telco_churn
WHERE internet_service <> 'No'
GROUP BY internet_service, tech_support
ORDER BY churn_rate_pct DESC;
-- Fiber + no support 49.4% | DSL + no support 27.8% | Fiber + support 22.6% | DSL + support 9.7%
```

Q5. Payment method and churn

```
SELECT payment_method, COUNT(*) AS customers,
       ROUND(100.0 * SUM(churn_flag) / COUNT(*), 1) AS churn_rate_pct
FROM telco_churn
GROUP BY payment_method
ORDER BY churn_rate_pct DESC;
-- Electronic check 45.3% | Mailed check 19.1% | Bank transfer 16.7% | Credit card 15.2%
```

Q6. Which segment should the retention team target first?

```
SELECT COUNT(*) AS customers, SUM(churn_flag) AS churned,
       ROUND(100.0 * SUM(churn_flag) / COUNT(*), 1) AS churn_rate_pct,
       ROUND(SUM(CASE WHEN churn_flag = 1 THEN monthly_charges ELSE 0 END), 0) AS mrr_lost
FROM telco_churn
WHERE contract = 'Month-to-month'
  AND internet_service = 'Fiber optic'
  AND tech_support = 'No';
-- 1,796 customers | 1,033 churned | 57.5% | 88,166 MRR lost
```

Q7. What share of all churn comes from each contract type?

```
WITH totals AS (SELECT SUM(churn_flag) AS all_churn FROM telco_churn)
SELECT contract, SUM(churn_flag) AS churned,
       ROUND(100.0 * SUM(churn_flag) / (SELECT all_churn FROM totals), 1) AS share_of_all_churn
FROM telco_churn
GROUP BY contract
ORDER BY churned DESC;
-- Month-to-month 88.6% | One year 8.9% | Two year 2.6%
```

5. Power BI Dashboard (step by step)

Step 1: Load the data

- Open Power BI Desktop, then **Home > Get data > Text/CSV** and select telco_clean.csv.
- Click **Transform Data**. Check types: tenure, monthly_charges, total_charges, churn_flag, tenure_order as numbers; the rest as text.
- Rename the query to telco and click **Close & Apply**.
- Alternative: **Get data > SQL Server / PostgreSQL** if you loaded the table into a database.

Step 2: Sort tenure groups correctly

- In Data view, click the **tenure_group** column, then **Column tools > Sort by column > tenure_order**.

Step 3: Create the measures

Home > New measure for each one below. Put them in a table called _Measures to keep the model tidy.

Total Customers = COUNTROWS(telco)

Churned Customers = CALCULATE(COUNTROWS(telco), telco[churn] = "Yes")

Churn Rate % = DIVIDE([Churned Customers], [Total Customers])

Total MRR = SUM(telco[monthly_charges])

MRR Lost = CALCULATE(SUM(telco[monthly_charges]), telco[churn] = "Yes")

MRR Lost % = DIVIDE([MRR Lost], [Total MRR])

Share of All Churn = DIVIDE([Churned Customers], CALCULATE([Churned Customers], ALL(telco)))

Churn Rate vs Average = [Churn Rate %] - CALCULATE([Churn Rate %], ALL(telco))

Avg Tenure Churned = CALCULATE(AVERAGE(telco[tenure]), telco[churn] = "Yes")

Avg Monthly Bill Churned = CALCULATE(AVERAGE(telco[monthly_charges]), telco[churn] = "Yes")

Format Churn Rate %, MRR Lost % and Share of All Churn as percentages, and MRR values as currency, using **Measure tools > Format**.

Step 4: Build the visuals

#	Visual	Fields	Purpose
1	4 KPI cards	Total Customers, Churn Rate %, MRR Lost, MRR Lost %	Headline numbers at the top
2	Clustered bar	Axis: contract Values: Churn Rate %	Contract effect
3	Column chart	Axis: tenure_group Values: Churn Rate %	Lifecycle drop-off
4	Clustered bar	Axis: internet_service Legend: tech_support Values: Churn Rate %	Service mix effect
5	Bar chart	Axis: payment_method Values: Churn Rate %	Payment risk
6	Donut	Legend: churn Values: Total Customers	Churned vs retained
7	Matrix (heatmap)	Rows: contract Columns: internet_service Values: Churn Rate %	Find the hot segment
8	Slicers	senior_citizen, gender, partner, dependents	Let viewers filter

Step 5: Add conditional formatting to the matrix

- Select the matrix, open **Format > Cell elements**, turn on **Background color**, click **fx**.
- Format style: Gradient, based on Churn Rate %. Minimum = light, Maximum = dark red. The 57.5% cell will stand out immediately.

Step 6: Layout and polish

- Top row: KPI cards. Middle row: contract, tenure, service mix. Bottom row: payment method and the heatmap. Slicers on the left.
- Use one colour theme (**View > Themes**). Use red only for churn. Turn off gridlines and add clear titles.
- Add a text box with the headline insight, for example: "Month-to-month fiber customers without tech support churn at 57.5%".
- Sync slicers across pages if you add a second page (**View > Sync slicers**).
- Save as churn_dashboard.pbix. Publish to Power BI Service (**Home > Publish**) if you want a shareable link, or export a screenshot and PDF for your portfolio.

6. Business Insights

#	Finding	Evidence
1	Churn is costly	26.5% of customers (1,869 of 7,043) have left, taking 139K of 456K MRR, which is 30.5% of recurring revenue.
2	Contract type is the biggest driver	Month-to-month churn is 42.7% vs 11.3% (one year) and 2.8% (two year). Month-to-month customers make up 88.6% of all churners.
3	The first year is the danger zone	47.4% of customers in their first 12 months churn, against 9.5% after 4 years. 1,037 of the 1,869 churners (55%) left within a year. Churned customers averaged 18 months of tenure vs 37.6 for retained.
4	Fiber customers leave far more	Fiber churn is 41.9% vs 19.0% for DSL and 7.4% for no internet. Fiber accounts for 1,297 of the 1,869 churners (69%). Worth checking pricing or service quality.
5	Support services protect revenue	Customers without tech support churn at 41.6% vs 15.2% with it. Without online security: 41.8% vs 14.6%.
6	Electronic check is a red flag	45.3% churn vs 15.2% to 19.1% for other payment methods. Automatic payments correlate with staying.
7	Higher-paying customers leave	Churned customers paid 74.44 per month on average vs 61.27 for retained ones, so the company loses its higher-value accounts.
8	Seniors are at higher risk	41.7% churn for senior citizens vs 23.6% for others.
9	One segment dominates	Month-to-month + fiber + no tech support: 1,796 customers (25.5% of base), 57.5% churn, 1,033 churners (55% of all churn), and 88K of the 139K lost MRR (63%).

Recommendations

- **Move month-to-month customers onto annual plans.** Offer a discount or a free month for switching. Contract length is the strongest lever in the data.
- **Bundle tech support and online security free for 3 to 6 months** for new fiber customers, and track whether churn drops.
- **Build a 90-day onboarding programme.** Check-in calls, setup help and a first-bill explanation for every new customer, because the first year is where more than half of churn happens.
- **Push automatic payment.** Offer a small bill credit for switching from electronic check to bank transfer or card.
- **Investigate fiber.** Compare fiber complaints, outage data and competitor pricing to find out why it churns twice as fast as DSL.
- **Assign the retention team the Q6 segment first.** It is a quarter of the base but more than half of the churn.

Illustrative impact: if retention actions saved just 10% of the churners in the top segment (about 103 customers), the company would keep roughly 8.8K in monthly revenue, or about 106K per year. This is an assumption to size the opportunity, not a forecast.

Limitations

- This analysis shows association, not cause. Fiber customers may churn for price or coverage reasons that this dataset does not contain.
- Payment method, paperless billing and contract type overlap with each other, so their effects are not independent.
- The dataset is a single snapshot, so trends over time cannot be measured. No prediction or scoring is attempted, as required.

Portfolio tip: put the clean.py script, SQL file, .pbix file, a dashboard screenshot and this summary in one GitHub repository with a README.